QUONTECH

ASSET MANAGEMENT

Asset Management Challenge 2026

Round 2 – A Tactical Deviation with Discretionary Equity Investing

Portfolio D – Inflation Resilient (Equal Weight)

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Group 8 | 7 November 2026; final data: 6 November 2026 closing prices

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1 Executive Summary and Mandate

Round 2 introduces a discretionary equity sleeve financed by selling part of the nine-ETF Strategic Asset Allocation (SAA) established in Round 1. The original equal-weight SAA remains the *unchanged benchmark*; the tactical portfolio must be long-only and have annualized ex-ante tracking error (TE) of 2–5% relative to that SAA. Risk is estimated from 60 aligned monthly USD returns. The challenge ends on 27 December 2026; the trades entered on 7–8 November use Friday 6 November closing prices.

Our three selected stocks express distinct theses: **Dollar General (DG)** – discount-retail recovery; **Chewy (CHWY)** – margin expansion despite slower sales; and **Micron (MU)** – AI-memory earnings growth and cyclical valuation. Target, Estée Lauder and Corcept were examined and rejected. In particular, a closer reading of the FDA materials invalidated the optimistic CORT approval case we had initially assumed; CHWY's peer multiple was independently rechecked and set at a more defensible, conservative level. Between the 30 Oct and 6 Nov pulls, MU fell ∼22% in a sector-wide memory sell-off (Kioxia's 2 Nov flat-guidance print); we responded by lowering MU's own valuation multiple, not by mechanically inflating its alpha off the lower price – see Section 2.3.

Stock	k (CAPM)	V_0	P_0	MoS	α (annual)	Within A
DG	9.60%	\$137.70	\$133.75	+2.9%	+3.2%	18.8%
CHWY	12.54%	\$26.56	\$22.56	+17.7%	+19.9%	58.7%
MU	14.18%	\$724.50	\$690.00	+5.0%	+5.7%	22.4%

Values use the confirmed 6 Nov 2026 Bloomberg closes. Weights within A are rounded.

The measured Treynor–Black unconstrained allocation is $w_A^* = 11.30\%$. A direct check on the model's continuous-weight sizing – the ex-ante tracking error estimated on the 12 final, whole-share position weights rather than on the smooth model weights – showed it running about 5bp above the model figure (4.05% vs 4.00% at a 4% model cap), so our internal TE cap is tightened to **3.90%** (Section 7) to keep the effective, post-rounding TE under 4% with a buffer. This implies $w_A = 10.69\%$; the remaining 89.31% stays in the ETF sleeve. The model gives TE 3.90%, ex-ante volatility 9.94% (SAA 8.63%) and expected Sharpe 0.93 (SAA 0.85). Post-execution, the full 12×12 covariance matrix confirms an effective ex-ante TE of 3.95% and an ex-ante volatility of 9.85% (model: 9.94%) on the actual final position weights (Section 7). These are *model estimates*, not realized forecasts for the short challenge window.

2 Selected Stocks: Valuation, Catalyst and Risks

2.1 Dollar General (DG)

Valuation. A dividend discount model is poorly suited to DG's modest payout; we use EPS-based relative valuation. Following DG's 27 August 2026 guidance of \$7.80–\$8.00 diluted EPS, we remove approximately \$0.25/share of non-recurring tariff-refund benefit: normalized EPS = **\$7.65**. We apply **18.0**× P/E, below direct peer Dollar Tree's approximately 19.47× forward P/E to reflect DG's management transition and recovery

risk. Therefore $V_0 = 18 \times \$7.65 = \$137.70$; against \$133.75 (6 Nov close), MoS is **+2.9%**. Source: DG Q2 FY2026 release/SEC 8-K (27 Aug 2026). This is a narrow margin that has compressed further as the price drifted up over the week; a lower justified multiple would already eliminate it.

Catalyst and risk. DG has now confirmed Q3 FY2026 results for **3 December 2026**, with a conference call the same day; Bloomberg's own expected-report-date field independently confirms 3 December. Confirmation of comparable-sales guidance (2.5–2.9%), earnings and inventory/margin improvement could support a re-rating before challenge-end. Risks are weak lower-income consumers, price competition and a slower-than-expected turnaround. CEO transition is expected in 2027, beyond the holding window.

2.2 Chewy (CHWY)

Method choice. A dividend-discount model is not appropriate for CHWY, which pays no regular dividend. Relative valuation lets us compare forward earnings against comparable companies on a like-for-like basis, so we use it as the primary method; a DCF is available as a cross-check below, but it depends on particularly sensitive assumptions about future margins and stock-based compensation, so we do not treat it as an independent confirmation.

Primary valuation – corrected comparable set. Forward EPS implied by CHWY's 12.84× forward P/E on the 5 Nov reference price (\$22.735) is \$1.7705, an estimate we treat as fixed (not re-derived daily from the live price) absent a fresh peer-multiple pull. A recheck of named e-commerce/pet-retail peers produced the following figures (source report's 6 Nov data pull):

Peer	Fwd P/E	Status	Peer	Fwd P/E	Status
ETSY	14.05×	Core	WOOF	10.28×	Core
EBAY	18.87×	Core	W	31.79×	Early-profit
REAL	52.00×	Early-profit			

Loss-making niche peers cannot supply meaningful forward P/Es. The median of the three sustainably profitable peers (ETSY, WOOF, EBAY) is **14.05**×; the five-name median is 18.87×, distorted by early-profitability observations. We apply 15×, a modest premium to the core median for CHWY's improving margins. Hence $V_0 = 15 \times \$1.7705 = \$26.56$; against the confirmed 6 Nov close of \$22.56, **MoS +17.7%**.

DCF cross-check. A five-year FCFE-style projection begins with FY2025 sales of \$12,601.5m and operating cash flow less capex of \$562.4m (4.46% margin). Revenue growth tapers from 7% to 5.5%, with FCF margin increasing from 5.2% to 8.0%. Discounting at $k = 12.54\%$ and terminal growth $g = 2.5\%$ gives **\$27.63/share**, or +22.5% versus the confirmed 6 Nov close. This depends materially on terminal margin and discount rate; the model uses diluted shares but does not forecast future stock-based-compensation dilution separately.

DCF (USD/share) $k \backslash g$	1.5%	2.0%	2.5%	3.0%	3.5%
11.5%	28.66	29.80	31.07	32.48	34.08
12.5% (base)	25.73	26.63	**27.62**	28.71	29.92
13.5%	23.49	24.22	25.02	25.89	26.86

Two adjustments follow from the latest Bloomberg financials. First, the equity bridge is updated to the 2 Aug 2026 balance sheet: cash and short-term investments of \$612.2m against \$588.7m of new long-term borrowings (operating leases excluded, as their cost is already in FCF), i.e. roughly zero net cash, and 401.5m shares after buybacks. Most of the cash reduction funded an acquisition of about \$553m (goodwill and intangibles rose from \$39m to \$484m) whose cash flows are not in the FY2025-based projection; treating it as fairly priced, the base value is essentially unchanged at ≈\$27.7. Second, operating cash flow adds back stock-based compensation (SBC) of \$297.9m in FY2025, 53% of FCF, while CHWY spent \$509.9m on buybacks over the last twelve months, largely to offset that dilution. Treating SBC as an economic cost lowers the DCF value materially:

SBC treatment	Value/share	vs \$22.56
Excluded (base case above)	\$27.7	+23%
Halved	\$23.2	+3%
Fading from 2.4% to 1.5% of sales	\$21.5	−5%
Constant at 2.4% of sales	\$18.8	−17%

Present value of SBC as a share of the same revenue path, at $k = 12.54\%$ and $g = 2.5\%$, deducted from the base DCF equity value; the full year-by-year revenue, FCF and SBC-path computation (not just this final figure) is reproducible in `Quontech_Code_Round2.py` (`chwy_dcf_fcfe`, `chwy_dcf_sbc_adjusted`).

Once SBC is costed, the DCF does not independently confirm the 15× valuation. We therefore report it as a sensitivity; the valuation rests on the peer multiple.

Catalyst and risk. Q3 results are expected on **10 December 2026** (Bloomberg expected report date). CHWY's slower sales growth prompted a sell-off, while adjusted EBITDA margin rose to 6.8% in Q2 FY2026 from 5.7% for FY2025. Stabilizing organic growth plus further margin expansion could prompt a positive repricing. Continued deceleration or weaker cash generation would instead validate a valuation discount; estimated residual volatility is 59.3% annually. Its revenue model is not directly comparable to marketplace peers, a limitation explicitly reflected in the sensitivity.

2.3 Micron Technology (MU)

Method choice. A dividend-discount model is likewise not suitable for MU, and a DCF is highly sensitive to the memory sector's cyclical earnings and cash flows – the same cyclicality that drove the Kioxia-triggered repricing below. We therefore apply a P/E multiple to a normalized EPS, cross-checked against sector peer multiples (the `PEERS_BBG` table below).

Valuation – revised after the Kioxia-driven sell-off. Between our 30 Oct and 6 Nov pulls, MU fell from \$884.74 to **\$690.00** (−22%), as part of a sector-wide memory

sell-off triggered by Kioxia's 2 Nov flat-guidance print (NAND ASP growth decelerating sharply) and comparable SK Hynix weakness. We deliberately do *not* plug the new price into the unchanged $V_0 = \$966$ (6.0×\$161): that would mechanically inflate MoS from +9.2% to roughly +40% on a week in which nothing about MU's own earnings power changed. The FY2027 normalized EPS is left at **\$161** (still the low end of the \$161–\$214.70 analyst range; if anything, most analysts have recently *raised* estimates, so the earnings outlook itself has not deteriorated). What we revise instead is the *multiple*, now grounded directly in a same-date, same-source Bloomberg pull (`PEERS_BBG`, 6 Nov 2026 close) rather than a secondary news report:

	Fwd P/E, 6 Nov	Fwd P/E, 30 Oct	Δ 1w
MU (consensus EPS basis)	3.93×	5.07×	−22.4%
SK Hynix	2.11×	2.68×	−21.1%
Kioxia	1.69×	2.70×	−37.4%

All three re-rated down together over the same week, confirming a sector-wide, not MU-specific, repricing. The comparison is still not perfectly like-for-like: fiscal year-ends differ (MU Aug, SK Hynix Dec, Kioxia Mar) and the multiples use each company's own consensus EPS, not a common definition. On MU's own consensus EPS (\$175.40), its current multiple is 3.93× – below our applied 4.5×. But on *our* more conservative, normalized EPS of \$161 (the basis this report actually uses), MU's current price already implies 690.00/161 = 4.29× – much closer to, and modestly below, the 4.5× we apply, which is the more relevant reconciliation for this report's own methodology. We apply **4.5**× – below the old 6.0× and below MU's own prior 5.3–6.8× range – reflecting the real, Bloomberg-confirmed multiple compression without assuming MU's memory mix is identical to SK Hynix's HBM-heavy one. $V_0 = 4.5 \times \$161 = \mathbf{\$724.50}$, yielding **MoS +5.0%** – smaller than the pre-sell-off +9.2%, not larger. The higher analyst consensus target is not used as intrinsic value. As a transparency check, not a change to the central case: at 4.0×, MoS is −6.7% (fails); at 4.5×, +5.0% (our case); at 5.0×, +16.7% – a reminder that the alpha is sensitive to a half-turn of multiple either side of our choice.

Catalyst and risk. Fiscal Q1 FY2027 results are expected on **17 December 2026** (Bloomberg expected report date), inside the challenge window. Stronger memory pricing, contracted HBM sales or higher earnings guidance could increase the multiple. The main value-trap risk is a sharp cycle reversal: supply growth could normalize pricing before the projected earnings materialize – the risk that, in our judgment, has already begun to show up in the price via the Kioxia-driven re-rating above, which is exactly why we lowered the multiple rather than taking the windfall gain. Estimated residual volatility is 65.2% annually. The unusually strong FY2026 margins in the original SEC-derived data should not be extrapolated as a sustainable run rate.

3 Rejected Candidates

Corcept Therapeutics (CORT). Initial modelling used a 65% probability of relacorilant approval on 17 Dec 2026. Re-examination of the FDA's corrected Complete Response Letter raised efficacy and liver-safety concerns (GRADIENT's missed primary

endpoint; highly enriched GRACE population; reported DILI signal). The publicly described resubmission focused on reanalysis of existing data; absence of a publicly described risk-mitigation plan is not proof none was submitted. We revised our *subjective* approval probability to 25–40%. Retaining the updated scenario inputs in the source report ($V_{\text{approval}} = \$115$, $V_{\text{rejection}} = \$52.76$, $P_0 = \$80$) produces:

Approval probability	25%	30%	35%	40%
Expected value (USD)	68.32	71.43	74.54	77.66
Estimated alpha	–15.6%	–11.4%	–7.3%	–3.1%

Even at 40%, the alpha is negative, so CORT is excluded rather than mechanically retaining a positive weight.

Estée Lauder (EL). An earlier margin-recovery thesis had already been reflected in the price by the time of our trading window. Repricing FY2027 EPS with a 29.24× L'Oréal peer multiple implies a negative margin of safety at the source report's reference prices; weight zero.

Target (TGT). After normalizing the one-off tariff-refund benefit, estimated EPS is \$8.71. A peer-multiple cross-check (rather than an assumed 18×) yields $V_0 \approx \$132.83$ versus a \$154.38 reference price, approximately –14% MoS. Differences between forward and trailing multiples are a limitation of the peer check; TGT is excluded under our conservative valuation.

4 Round 1 Realized Performance

Pulled from `ROUND2_BBG` (total-return-adjusted series, 15 trading days to the 6 Nov 2026 close):

Metric	Realized	Metric	Mandate
Cumulative return (TR)	−2.00%	Volatility target	8% (7–9%)
Annualized realized volatility	7.1%	Sharpe (realized)	−4.63

Two distinct NAV figures appear in `ROUND2_BBG` and must not be conflated: the *total-return* NAV – capital × (1 + cumulative TR return) – is **\$98.00m** (\$98,001,010, 6 Nov); the *price-only* position value (shares × PX_LAST, using this script's own buy-and-hold share-count estimate) is \$97,904,979 (≈ \$97.90m), about \$96,031 lower. This is now settled: a live Bloomberg `PRTU` position snapshot for `TRI-FORCE` (7 Nov 2026) reports an absolute Total market value of **\$98,000,724.83** – confirming the TR-basis \$98.00m figure directly, not derived. The gap is *not* uninvested cash: the snapshot's actual cash is only \$1,134.93 (0.0012% of NAV). Comparing the snapshot's real share counts to this script's buy-and-hold estimate shows IVV/VEA/XLV/XLE/GLD/PDBC match (to fractional-share rounding), while TIP/SHY/IGSB hold materially *more* shares (+158/+425/+855) than the estimate – those three bond ETFs' distributions were reinvested into additional shares (DRIP), not paid out as cash, which (together with the \$1,135 of real cash) accounts for essentially the whole \$96,031 gap. The correct Round

2 sizing basis is **$98.00m**, now confirmed by Bloomberg's own number as well as the TR-return calculation, used below in place of the illustrative $100m.

Performance attribution. Initial-weight approximation $w_{j,0}R_j$ (realized return contribution, not prior risk contribution), all 9 ETFs, from `ROUND2_BBG` columns U–W:

ETF	Return	Contrib.	ETF	Return	Contrib.
IVV	−3.7%	−0.41%	TIP	+0.8%	+0.09%
VEA	−2.3%	−0.25%	SHY	+0.6%	+0.07%
XLV	+4.6%	+0.51%	IGSB	+0.7%	+0.08%
XLE	−9.7%	−1.08%	GLD	−2.5%	−0.28%
PDBC	−6.7%	−0.74%			

XLE and PDBC drove essentially all of the drawdown (together −1.82pp of the −2.00% total); XLV was the only meaningfully positive contributor. The realized volatility and Sharpe computed on 15 daily returns are noisy relative to annualized ex-ante SAA estimates – the −4.63 realized Sharpe in particular reflects a short, largely energy/commodity-driven drawdown, not a forecast; report the observation without over-interpreting it.

5 Checkpoint 1 – Alphas and Portfolio Construction

Stock-specific expected alphas use the course convention $\alpha_i = (1+k_i)(V_i/P_i - 1)$. They are model-based annualized values derived from valuation assumptions; they do not measure realized challenge-period alpha. Negative-alpha candidates are rejected (Section 3).

Stock	α_i	β_i vs SAA	$\sigma(\varepsilon_i)$	IR_i	Weight in A
DG	+3.2%	0.83	34.9%	0.09	18.8%
CHWY	+19.9%	1.39	59.3%	0.34	58.7%
MU	+5.7%	2.65	65.2%	0.09	22.4%

Rather than assuming independent residuals, we use the measured 60-month residual correlation matrix, rolled forward to the window Nov 2021–Oct 2026 ($\rho_{DG,CHWY} = 0.043$, $\rho_{DG,MU} = 0.056$, $\rho_{CHWY,MU} = -0.189$): $w_A^{\text{stocks}} \propto \Sigma_\varepsilon^{-1}\boldsymbol{\alpha}$. The active-sleeve information ratio is

$$IR_A = \sqrt{\boldsymbol{\alpha}^\top \Sigma_\varepsilon^{-1} \boldsymbol{\alpha}} = \mathbf{0.376}$$

(diagonal-only: 0.360); it is sensitive to the CHWY valuation, and is *not* evidence of realized manager skill. When residual returns are uncorrelated, our covariance-matrix approach reduces to the standard Treynor–Black weights, proportional to alpha divided by residual variance. CHWY now carries 58.7% of the active sleeve (up from 49.0% pre-sell-off) purely because DG's and MU's alphas shrank while CHWY's widened slightly – a concentration that deserves the same scrutiny flagged for CHWY's valuation elsewhere in this report.

Fundamental Law check (implied IC, not a measured one). $IR_A \approx IC\sqrt{\text{breadth}}$ (Grinold–Kahn). Treating our three independent stock calls as breadth = 3 – a simplifying assumption, not an empirically estimated breadth – backs out an *implied* average

information coefficient $IC \approx 0.376/\sqrt{3} = 0.217$ per call (diagonal case: $0.360/\sqrt{3} = 0.208$) consistent with the model's own IR_A. This is a plausibility check on the model, not an observed measure of our forecasting skill: we have no track record of resolved calls to estimate IC from directly. An implied IC above ~ 0.10–0.15 on single-name valuation judgments would be a demanding level of skill to actually possess with only 3 bets – a further reason to read $IR_A = 0.376$ as a model output contingent on our valuation calls holding up, not as an already-demonstrated edge.

The underlying monthly regression data and reproducible calculation are held in `ROUND2_BBG` and `Quontech_Code_Round2.py`. The 60-month regression window (beta, $\sigma(\varepsilon)$, residual correlations above) was rolled forward to Nov 2021–Oct 2026 on 7 Nov 2026 (October's month-end close appended to `ROUND2_BBG` and the OLS regressions re-run), satisfying the mandate's "latest 60 months" rule.

6 Checkpoint 2 – Value-Trap and Quality Checks

Company	ROE	FCF margin	Debt/equity	Revenue growth	Gross margin	Capital return
DG	19.7%	3.3%	0.49×	+4.3%	32.6%	Div./FCF ≈35%
CHWY	N/A[1]	4.46%	N/A[1]	+7.3%	N/A[2]	Not material
MU	61.4%	44.3%	0.04×	+256.4%	80.7%	Low

Periods and limitations: DG uses a mixture of FY2026 H1 and TTM figures (company 10-Q, July 2026); including leases, DG debt/equity is 1.70×. CHWY uses FY2025 FCF and Q2 FY2026 sales growth (company 8-K/10-Q); [1]equity is negative so ROE/debt-to-equity is not meaningful; [2]Q2 adjusted EBITDA margin is 6.8% against 5.7% FY2025. MU uses FY2026 results (company 8-K/10-K); peak-cycle profitability must not be extrapolated. Cross-company comparisons are indicative because reporting periods differ.

Peer and valuation checks. DG's selected 18× P/E is below DLTR's approximate 19.47×; DLTR's H1 FCF margin is around 10.8% versus DG 3.3%. CHWY's core peer forward-P/E median is 14.05×, with two early-profitability names excluded from the core set. MU's multiple was revised from 6× to 4.5× on 6 Nov (Section 2.3), cross-checked against Bloomberg's own same-date `PEERS_BBG` pull (MU 3.93×, SK Hynix 2.11×, Kioxia 1.69× consensus forward P/E, all down 21–37% over the prior week), though fiscal periods across the three still differ. As additional checks, a broader Bloomberg blended approach in the source report implies values of \$167.23 (DG) and \$1,111.35 (MU), but uses different peer groups and should not replace the deliberately conservative base valuations. These checks support further diligence, not certainty that the stocks are free of value traps.

Value-trap interpretation. *DG:* the thesis rests on margin recovery and management's ability to execute on operating efficiency; with a margin of safety of only +2.9% (Section 2.1), there is little room for error if that recovery is slower or weaker than assumed. *CHWY:* FCF growth must be read together with stock-based compensation and buybacks (Section 2.2) – the risk is that an accounting-level improvement in margins does not translate into an equivalent increase in value for shareholders once dilution is properly costed. *MU:* current profitability may reflect a cycle peak rather than a

sustainable run rate (Section 2.3); the risk is that the normalized EPS we use stays too high relative to what the memory cycle can sustainably deliver over the long run. These conclusions link the quantitative indicators above to the possibility that an apparently undervalued stock is in fact a value trap, rather than leaving that inference implicit.

7 Tracking-Error Budget and Portfolio Risk

The active sleeve has $\alpha_A = 13.61\%$, $\beta_A = 1.5689$ and $\sigma(\varepsilon_A) = 36.17\%$ from 60 aligned monthly observations (window Nov 2021–Oct 2026). The SAA benchmark itself keeps its Round 1 strategic equal weights, but its *risk* parameters are re-measured on the same rolled-forward window (not left at the Round 1 report's Sep2020-Aug2025 figures): $E(Re_S) = 7.31\%$ and $\sigma_S = 8.63\%$ (both from `ROUND2_BBG`'s equal-weight SAA return series, Nov 2021–Oct 2026). The model calculates:

$$w_{0,A} = \frac{\alpha_A/\sigma^2(\varepsilon_A)}{E(Re_S)/\sigma_S^2} = 10.61\%,$$
$$w_A^* = \frac{w_{0,A}}{1 + (1-\beta_A)w_{0,A}} = \boxed{11.30\%}.$$

The TE per unit of active sleeve is 36.50% (up from 33.12% pre-sell-off, since MU's price drop raises its weight in the risk calculation). The full 2–5% TE mandate therefore permits approximately 5.48–13.70% active weight.

Theoretical vs. effective ex-ante TE. Sizing to a 4% *model* cap gives $w_A = 4\%/36.50\% = 10.96\%$. But the model's TE formula assumes continuous weights; the effective ex-ante TE – estimated on the 12 *final, whole-share* position weights, computed directly from those weights and the 60 monthly historical returns, not from the reduced-form $\alpha_A/\beta_A/\sigma(\varepsilon_A)$ model – came out at **4.05%**, about 5bp above the model figure. The dominant cause is methodological, not rounding: the reduced-form model assumes the passive sleeve's return moves exactly with the equal-weight strategic SAA benchmark, but we sell w_A of each ETF's *current, price-drifted* share count (Section 4), so the remaining passive sleeve keeps those drifted weights, not the strategic equal weights – a mismatch the model cannot see. Whole-share rounding on top of that is a secondary, smaller effect. That is still inside the official 2–5% mandate but marginally breaches our own internal cap, an inconsistency worth removing rather than carrying into execution. We therefore tighten the **internal safety cap to 3.90%** (model TE) and resize to $w_A = 3.90\%/36.50\% = \boxed{10.69\%}$. After the trades were entered in `PRTU` and confirmed in `PORT` to match the 12 final position weights exactly, we re-verified the effective ex-ante TE directly: building the full 12×12 sample covariance matrix from the historical monthly returns of all 9 ETFs and the 3 stocks (same Nov 2021–Oct 2026 window, 60 observations) and evaluating it at the real final weights gives

$$TE = \sqrt{(\mathbf{w}_P - \mathbf{w}_{SAA})^\top \, \Sigma \, (\mathbf{w}_P - \mathbf{w}_{SAA}) \times 12} = \mathbf{3.95}\%$$

– below the 4% cap, confirming the buffer worked. The same full matrix gives an ex-ante

portfolio volatility of **9.85%**, against 9.94% under the reduced-form model; the two are not inconsistent, they come from slightly different methods (full 12-asset covariance vs. the $\alpha_A/\beta_A/\sigma(\varepsilon_A)$ reduction), and the parametric 1-month 5% VaR implied by 9.85% is **4.68%**, against 4.72% under the reduced-form figure, under the same assumptions. The remaining 89.31% stays in the passive ETFs.

Risk measure	SAA	Tactical (model)	Tactical (full matrix)
Ex-ante volatility	8.63%	9.94%	9.85%
1-month 5% VaR	4.10%	4.72%	4.68%
Tracking error vs SAA	–	3.90%	3.95%
Exp. excess return (ann.)	7.31%	9.20%	–
Expected Sharpe	0.85	0.93	–

"Model" is the reduced-form $\alpha_A/\beta_A/\sigma(\varepsilon_A)$ sizing formula (continuous weights, passive sleeve assumed to track the SAA benchmark exactly); "full matrix" is the 12×12 sample covariance of actual monthly returns evaluated at the real final position weights (Section 7). The two differ mainly because the real passive sleeve keeps its price-drifted ETF weights, not the SAA's strategic equal weights – not primarily because of whole-share rounding.

The 7–9% target from Round 1 applies to the *unchanged SAA*, not the new tactical portfolio. The tactical portfolio's expected Sharpe improvement is sensitive to the three alphas, especially CHWY. Monthly covariance-based TE does not eliminate the possibility of adverse single-day catalyst surprises.

8 Portfolio Description and Execution

Sizing below uses the actual Total market value from a live Bloomberg `PRTU` position snapshot for `TRI-FORCE`, dated 7 Nov 2026 ($98,000,724.83, Section 4), and that same snapshot's *exact* current share counts per ETF – not an equal-weight or drifted-weight estimate. This is an executable trade list, sized at the revised $w_A = 10.69\%$ (Section 7).

ETF	Current shares	SELL	Remaining
IVV	14,492.000	1,548.000	12,944.000
VEA	159,895.000	17,085.000	142,810.000
XLV	62,877.000	6,718.000	56,159.000
XLE	162,824.000	17,398.000	145,426.000
TIP	105,437.107	11,266.369	94,170.738
SHY	136,992.290	14,638.165	122,354.125
IGSB	217,024.100	23,189.878	193,834.222
GLD	29,085.000	3,107.000	25,978.000
PDBC	552,516.000	59,038.000	493,478.000

Stock	BUY (shares)	Actual cost (USD)
DG	14,753	$1,973,214
CHWY	272,671	$6,151,458
MU	3,401	$2,346,690

All securities are USD-denominated. ETF sales are $w_A = 10.69\%$ of each ETF's *actual current* share count (Bloomberg PRTU, 7 Nov snapshot) – whole shares for IVV/VEA/XLV/XLE/GLD/PDBC, fractional for TIP/SHY/IGSB since those carry fractional holdings already (DRIP). The stock column reports *actual cost* (whole shares × 6 Nov close), not the pre-rounding dollar budget, which runs $401 higher across the three names and is not the cash figure that matters for reconciliation. Total ETF sale proceeds: $10,470,755, plus opening cash of $1,134.93, against total actual stock purchase cost of $10,471,361.51 – a final cash residual of **$528.07**. This table is reconciled directly against Bloomberg's own position file, not reconstructed from an NAV/weight estimate.

Final post-trade portfolio (all 12 positions). For completeness, full names, 6 Nov closing prices and resulting weights:

Tkr	Name	Price	Final wt
IVV	iShares Core S&P 500 ETF	$738.68	9.76%
VEA	Vanguard FTSE Developed Mkts ETF	$67.90	9.89%
XLV	Health Care Sel. Sector SPDR	$184.80	10.59%
XLE	Energy Sel. Sector SPDR	$61.62	9.14%
TIP	iShares TIPS Bond ETF	$106.27	10.21%
SHY	iShares 1–3Y Treasury Bond ETF	$81.63	10.19%
IGSB	iShares 1–5Y IG Corp. Bond ETF	$51.57	10.20%
GLD	SPDR Gold Shares	$372.52	9.87%
PDBC	Invesco Opt. Yield Div. Commodity ETF	$18.77	9.45%
DG	Dollar General Corp.	$133.75	2.01%
CHWY	Chewy, Inc. (Class A)	$22.56	6.28%
MU	Micron Technology, Inc.	$690.00	2.39%
Cash	*USD residual*	–	0.0005%

Weights sum to 100.0000% of the $98,000,724.83 NAV (reconstructed to the cent). Active sleeve (DG+CHWY+MU) = 10.69%, matching w_A above.

Execution rule. On 7–8 November, modify the *existing* Round 1 portfolio in Bloomberg `PRTU` using the trade list above: sell $w_A = 10.69\%$ of each ETF's actual current share count, then buy DG, CHWY and MU with the proceeds in the amounts shown. This has been done: the 12 resulting positions and the $528.07 cash residual were confirmed in `PORT` to match this table exactly, and the effective ex-ante tracking error on those final quantities was independently verified at 3.95% (Section 7).

Reproducibility and final checks. All three 6 Nov closing prices, the 60-month regression window (rolled forward to Nov 2021–Oct 2026, for both the three stocks and the SAA's own risk parameters), and the real Bloomberg PRTU position snapshot (NAV and exact share counts, 7 Nov) are pulled into `ROUND2_BBG/PEERS_BBG` and re-run through `Quontech_Code_Round2.py`; every number in this report, including the trade list above, reflects that run. All three stocks' upcoming earnings dates (DG 3 Dec, CHWY 10 Dec, MU 17 Dec) are independently confirmed by Bloomberg's own expected-report-date field.